

Model Settings and Output Files

Documentation Overview

August 18, 2026

Model version: 1.0.3

This document provides an overview of the key configuration parameters and output files used in the energy system optimization model FEM.

1 Model Settings

All settings in the model have a sensible default value. Any of these defaults can be overridden by the user by adding the corresponding setting item to the scenario setup CSV which is the file referenced by the *target_csv* variable in *scenarios.py*.

To help users, we distinguish between two groups: primary settings and secondary settings. Primary settings are the parameters commonly modified through the standard settings file; they are intended to be easy to change and typically do not require deep knowledge of the model. Secondary settings are more advanced parameters that also have defaults but are usually only added to the settings file when needed — overriding them may require additional code changes, extra input files, or further testing.

1.1 Primary settings

The settings file to be used is defined in *scenarios.py* at the definition of the variable *target_csv*. If those settings are removed from the file, the default value is used.

Setting	Description
<i>Item</i>	Scenario's and resulting output folder's name.
<i>sub_secn</i>	Subscenario's name. If a scenario is supposed to have several subscenarios, <i>Item</i> must be the same but <i>sub_secn</i> must be different.
<i>weight_in_objective_fcn</i>	The weight of each subscenario. It must add up to 1 across all subscenarios for each scenario.
<i>NTC_CH_ratio</i>	Scaling factor for the Swiss NTC. Default: 1.
<i>limit_fuel_import_CH</i>	Scaling factor for the upper limit of the Swiss fuel import according to the contents of the file <i>fuel_limits.csv</i> . If no limit is desired, it can be set to "False". Default: False.
<i>weather_year</i>	Weather year used for demand and renewable generation profiles. Available years are 1995, 2008 and 2009. Default: 1995.

Setting	Description
run_year	Target year of the model run in terms of the renewable expansion, etc. Available years are 2035 and 2050. Default: 2050.
t_start	Starting hour of the simulation. Hour 6553 corresponds to the beginning of October. Starting here aligns with modeling a hydro year. Default: 6553.
t_end	Last hour of the simulation. If $t_{start} > t_{end}$, the simulation will continue with t_1 after t_8760. Default: 6552.
electrolyzer_demand _reduction_coefficient	Scaling factor that modifies the annual electricity demand of all electrolyzers but the Swiss ones. These demands can be found in <i>electrolyzer_net_demand.csv</i> . Default: 1.
flexible_household _heatpump_share	Share of households that operate their heat pumps in a flexible way. Default: 0.5.
share_of_flexibly _charging_EV	Share of the total EV consumption that is charged flexibly. The remaining share follows a fixed charging profile. Default: 0.5.
V2G_share_of_flexibly _charging_EV	Share of the flexibly charging EVs that do bidirectional charging. Default: 0.5.
reduce_inflex_demand_by _[MWh]	Inflexible demand reduction applied to CH. A negative value can increase the demand. Default: 0.
reduce_DH_demand_by _[MWh]	District heating demand reduction applied. A negative value can increase the demand. Default: 0.
ror_annual_TWh	The annual ror generation is scaled to this value. If "False", the raw data is not scaled. Default: False.
hydro_inflow_TWh	The annual open psp generation is scaled to this value. If "False", the raw data is not scaled. Default: False.
pump_capacity_GW	Controls how much power the large Swiss psp should have. If "False", the raw data is not scaled. Default: False.
adding_hydro_storage _cap_TWh	Additional hydro storage capacity in TWh. If "False", nothing is added. Default: False.
heat_flexibility_Kelvin	Thermal flexibility range in K. The temperature in flexibly heating households can deviate this much from 22°C. Default: 0.5
resistive_heater _investment_cap_MW_total	Upper limit in MW_{th} for the total resistive heater investment summed over all district heating nodes. If set to "False", no cap is applied. Default: False.

Setting	Description
<code>PVRF_CH_coefficient</code>	Scaling factor for the already available PV capacity in Switzerland prior to any investments. If 1, the PV capacity in 2035 is 30.1 GW and in 2050, it is 35.5 GW. Default: 1.
<code>battery_cost_factor</code>	Multiplier applied to battery investment costs. If 1 and if the costs have not been modified in the <i>input\cost_operation_invest_data.py</i> file, the investment costs in power capacity will be 44'696 CHF/MW and 38'312 CHF/MW in 2035 and 2050 respectively. The investments costs in storage capacity will be 51'581 CHF/MWh and 44'212 CHF/MWh. Default: 1.
<code>battery_investment_nodes_in_addition_to_CH</code>	Comma-separated list of nodes (e.g. "DE00, FR00") in which the model can invest in large-scale batteries in addition to Switzerland. If empty, battery investments are only possible in Switzerland. Default: empty.

1.2 Secondary settings

Those settings can be added manually to the settings file to change them.

Parameter	Description / Default Value
<code>solver_name</code>	Optimization solver used. Default: gurobi.
<code>allow_investment</code>	Enables investments. Default: True.
<code>merge_some_countries</code>	Merges smaller countries for simplification. Default: True.
<code>minimum_RES_target_CH</code>	If set to an annual energy amount in MWh, the renewable generation in CH (existing infeed plus potential generation of newly invested plants) must reach at least this value. Default: False.
<code>rep_hydro_plants</code>	If True, the large Swiss pumped storage and open hydro plants are represented by three representative generators instead of one. Default: True.
<code>cost_op_match_nexuse</code>	If True, the operation costs are overwritten with the cost assumptions of the Nexus-e model for model comparison purposes. Default: False.
<code>slack_soc</code>	If True, slack variables with a high penalty are added to the storage balance constraints (debugging aid). Default: False.

Parameter	Description / Default Value
high_resolution_PV	If True, the seven Swiss PV investment candidates are replaced with per-plant rooftop PV candidates based on Nexus-e data read from <i>input\RES_nexuse\PV</i> . The standard PV profiles are still used for the pre-existing capacity. Default: False.
DH_demand_adjustment_file	Either "False" (off) or the filename of a CSV in <i>input\demand\adjustments</i> holding signed MW deltas (district heating node $\times$ hour) that are added to the assembled district heating demand. Default: False.
CH_only	Runs the model in a Switzerland-only configuration that belongs to a separate model variant. Not used in standard runs. Default: False.
neighbor_countries_for_CH_only_mode	Neighboring countries considered in the Switzerland-only configuration. Only relevant if <i>CH_only</i> is enabled.
neighbor_price_scenario_for_CH_only_mode	Price scenario used in the Switzerland-only configuration. Only relevant if <i>CH_only</i> is enabled.
RES_CH_coefficient	Scaling factor for renewables in CH before any investments. The unscaled data is saved in <i>input\res_capacities_CH.csv</i> . It has been derived from the original policy dependent data from the Energy Perspectives 2050+ saved in <i>input\res_capacities_CH_old.csv</i> . Default: 1.
Windon_CH_coefficient	Scaling factor for onshore wind in CH before any investments. The unscaled data is saved in <i>input\res_capacities_CH.csv</i> . It has been derived from the original policy dependent data from the Energy Perspectives 2050+ saved in <i>input\res_capacities_CH_old.csv</i> . Default: 1.
share_of_available_charging_capacity_for_V2G	The share of their charging capacity that EV owners make available for V2G. They are expected not to provide 100% to conserve their batteries. Default: 0.7.
storage_to_charge_ratio_days	To avoid unrealistic behavior, PTES cannot charge or discharge entirely in less than the given amount of days. If set to 0, the constraint is effectively disabled. Default: 7.
lost_load_cost_mode	Cost modeling of lost load. It can be "single_step" or "multi_step_v1". In case of the latter one, the lost load steps can be defined in <i>input\LostLoadCost\lostLoadCost_multi_step_v1.csv</i> . Default: single_step.

Parameter	Description / Default Value
Node_list_setting	List of modeled nodes. Default: ["CH00", "IT00", "DE00", "FR00", "AT00", "DKE1", "DKW1", "BE00", "CZ00", "ES00", "UK00", "HU00", "LU00", "NL00", "PL00", "PT00", "SI00", "SK00", "HR00", "SE01", "SE02", "SE03", "SE04", "NON1", "NOM1", "NOS0",]
NodeDH_list	Set of district heating nodes in the model. It can be set to zero when entering an empty list, i.e. [] For the default, please refer to the <i>scenarios/settings-default.py</i> file.
eu_policy	EU policy setting. "DE" (Distributed Energy) and "GA" (Global Ambition) are available. Default: "GA".
ch_policy	Swiss energy policy configuration from the Energy Perspectives 2050+; ZBA = Zero BAis. Most of the data is also available for "WWB", "ZVA" and "ZVB" but currently, only "ZBA" is available. Default: "ZBA".
winter_limit	A certain net import limit that cannot be exceeded in a given timeframe. If the mode is False, this constraint is not active. Default: {"mode": False, "window": [6553, 2184], "energy_MWh": 5 * 1000000}.
limited_fuels_import.CH_list	List with the fuels with a restricted import, if enabled. Default: ["resmethane", "fossilmethane"].
RES_EU_coefficient	Scaling factor for the non-CH renewable capacities. Default: 1.1.

2 Model Output Files

The model produces several CSV output files and one log file summarizing generation, storage, investments, and system balances. All files of a run are written to the folder *output/[Scenario Name]* defined in the scenario setup CSV. The following overview lists the key outputs, their content, and units. Please note that all files end with *.csv*.

Each file is marked with either **i** for input or **o** for output to indicate whether it serves as an input to the model—typically a parameter—or as an output from the model—typically a variable.

File Name (Input/Output)	Content	Unit
avail_plant(i)	Hourly max utilization for each plant	
BA_max_heating_capacity(i)	Max heating capacity of the heat pumps	MW

File Name (Input/Output)	Content	Unit
BA_th_con(i)	Building archetype hourly thermal consumption	MWh
BA_th_lim(i)	Thermal limit per building archetype; the building archetype thermal limit represents how much the building can be heated or cooled down stepping away from the comfort temperature allowing for the use of thermal inertia	MWh
constraint_scaling(i)	Per-constraint row scaling factors used to improve the numerical conditioning of the linear program; solution neutral	
Constraint_investment_res_CH_dual(o)	Dual value of the minimum renewable generation constraint for CH; only written if <i>minimum.RES.target.CH</i> is active	CHF/MWh
Constraint_winter_limit_dual(o)	Dual value of the winter net import limit constraint; only written if <i>winter_limit</i> is active	CHF/MWh
COP(i)	Coefficient of performance	
cost_data_inv_discharge_slp(o)	Cost of investing in discharge capacity for each technology; only hydrogen has this because the facilities for generating hydrogen and storing hydrogen are considered separately	CHF/MW
cost_inv_dict(o)	Total investment costs of non-thermal plants	CHF
cost_inv_fuel_storage_dict(o)	Total investment costs of fuel storages	CHF
cost_inv_thermal_dict(o)	Total investment costs of thermal plants	CHF
cost_op_dict(o)	Annual operation costs of the individual plants	CHF
cost_op_thermal_dict(o)	Annual operation costs of the individual thermal plants	CHF
curtailment(o)	Occurred curtailment	MWh
curtailmentTh(o)	Occurred thermal curtailment	MWh _{th}
Data_plant_flex_d_within_window(i)	Flexibility data of the electrolyzers: time windows and the electricity consumption fixed within each window	MWh
demand(i)	Fixed demand	MW
demandDH(i)	Fixed district heating demand	MW _{th}

File Name (Input/Output)	Content	Unit
dsrThDev(o)	Thermal demand side response; records the deviation of the representative building's thermal energy level from the target level for buildings heated by district heating	MWh _{th}
dsrThDev_max(i)	Max thermal demand side response	MWh _{th}
emission_factor_per_MWh(i)	CO ₂ emission factor per plant, expressed per MWh of electricity output and thus accounting for the plant efficiency	tCO ₂ /MWh
emissions_dict(o)	Annual CO ₂ emissions per plant and subscenario, computed from the generation and the emission factor; for reporting only, emissions do not enter the objective function	tCO ₂
energy_balance_dual(o)	Dual value of the electric energy balance constraint; electricity price	CHF/MWh
energy_balancethermal_dual(o)	Dual value of the thermal energy balance constraint; heat price	CHF/MWh _{th}
energy_max_limit(i)	Max energy capacity that can be invested for certain plants in CH	MWh
energyTH_max_limit(i)	Max energy capacity of thermal energy storage assets that can be invested in CH	MWh _{th}
EV_inflexible_demand(i)	Fixed charging profile of the inflexibly charging share of the EVs per node and hour; see <i>share_of_flexibly_charging_EV</i>	MWh
Export(o)	Utilization of each transmission line	MW
exported(-)	unused	MWh
fuel_consumption_of_fuel(o)	Consumed fuel	MWh
fuel_consumption_of_plant(o)	Consumption per plant	MWh
fuel_import_capacity_annual(i)	Annual fuel import capacity	MWh
fuel_production_capacity_CH_annual(i)	Annual fuel production capacity	MWh
fuel_storage_capacity_annual(o)	Invested storage capacity of fuel storage	MWh
fuel_storage_investment_annual_limit(i)	Max investment allowed in fuel storage capacity	MWh
gen(o)	Hourly generation per plant	MWh

File Name (Input/Output)	Content	Unit
<code>gen_energy_max(o)</code>	Energy capacity of storage plants. For plants without investment option it is fixed to the existing capacity	MWh
<code>gen_energy_max_reduced_cost(o)</code>	Reduced costs of the storage energy capacity investments: how much cheaper the energy capacity would need to be to become competitive	CHF/MWh
<code>gen_energyTh_max(o)</code>	Thermal energy capacity of thermal storage plants. For plants without investment option it is fixed to the existing capacity	MWh _{th}
<code>gen_energyTh_max_reduced_cost(o)</code>	Reduced costs of the thermal storage energy capacity investments	CHF/MWh _{th}
<code>gen_max(o)</code>	Max generation capacity/ Installed capacity; do not confuse with <i>gen_max_limit</i> which is a parameter whereas <i>gen_max</i> is a variable to the model	MW
<code>gen_max_infeedp(i)</code>	Max generation capacity of pre-existing RES plants	MW
<code>gen_max_limit(i)</code>	Upper limit of <i>gen_max</i>	MW
<code>gen_max_reduced_cost(o)</code>	Reduced costs of the generation capacity investments	CHF/MW
<code>generationTh_limit_dual(o)</code>	Dual value of the thermal generation capacity limit; scarcity rent of the thermal generation capacity of heat pumps, resistive heaters and thermal storages	CHF/MW _{th}
<code>genTh(o)</code>	Hourly thermal generation per plant	MWh _{th}
<code>genTh_max(o)</code>	Max thermal generation capacity/ Installed capacity; do not confuse with <i>genTh_max_limit</i> which is a parameter whereas <i>genTh_max</i> is a variable	MW _{th}
<code>genTh_max_limit(i)</code>	Upper limit of <i>genTh_max</i>	MW _{th}
<code>genTh_max_reduced_cost(o)</code>	Reduced costs of the thermal generation capacity investments	CHF/MW _{th}
<code>HP_inflexible_demand(i)</code>	Fixed consumption profile of the households that do not operate their heat pumps flexibly, per node and hour; see <i>flexible_household_heatpump_share</i>	MWh
<code>imported(-)</code>	unused	
<code>infeed(i)</code>	Hourly infeed from pre-existing RES	MWh
<code>inflow(i)</code>	Natural inflow to storage plants	MWh
<code>investment_emax_slp(i)</code>	Storage investment costs	CHF/MWh

File Name (Input/Output)	Content	Unit
investment_emax_slpTh(i)	Thermal storage investment costs	CHF/MWh _{th}
investment_fuel_storage_slp(i)	Fuel storage capacity investment costs	CHF/MWh
investment_genmax_slp(i)	Generation capacity investment costs	CHF/MW
investment_genmax_slpTh(i)	Thermal generation capacity investment costs	CHF/MW
investment_summary(o)	Automatically generated investment overview	
large_objective_coefficients(o)	Diagnostic list of variables with objective coefficients above 10 ⁶ ; only written if such coefficients are found	
line_trade_price (i)	Only used in a Switzerland-only configuration that belongs to a separate model variant; contains zeros in standard runs and can be ignored	
lostload(o)	Hourly lost load	MWh
lostload_capacity_per_step(i)	Lost load per lost load step; related to <i>lost_load_cost_mode</i>	MWh
lostload_cost_dict(o)	Total lost load costs per subscenario	CHF
lostload_cost_per_step(i)	Lost load costs per lost load step; related to <i>lost_load_cost_mode</i>	CHF/MWh
Map_[a]_[b] (i)	Internally used mapping from a to b	
operation_qdr(i)	Factor of the former quadratic term of the operation costs. The quadratic term is disabled since model version 1.0.2, so the file is exported but has no effect on the optimization	(CHF/MW) ²
operation_slp(i)	Factor in the operation costs' linear term of the objective function	CHF/MW
operation_slpTh(i)	Factor of the thermal operation costs of the objective function	CHF/MW _{th}
outflow(i)	Naturally hourly lost electric energy and consumption of EVs	MWh
P_allinv(i)	All plants/planttypes that the model can invest in	
pmp_max(i)	Charging limit of storages	MW
pumpTh_max(i)	Charging limit of thermal storages	MW _{th}
settings(i)	Settings that were used during this model's execution	
soc(o)	Storages' State of charge	MWh

File Name (Input/Output)	Content	Unit
socTh(o)	Thermal storages' State of charge	MWh _{th}
spill_water(o)	Hourly curtailed hydro inflow	MWh
statistics(o)	Solver termination information	
storage_charge(o)	Hourly charging of storages	MWh
storage_charge_eff_in (i)	Efficiency when charging storages	
storage_charge_eff_out (i)	Efficiency when discharging storages	
storage_chargeTh(o)	Hourly charging of thermal storages	MWh _{th}
storage_soc_dual(o)	Dual variable of the storage balance constraint; water value / storage value	CHF/MWh
storage_start_cond(i)	Relative level at which storages start and end	
storageTh_rate_limit _dual(o)	Dual value of the thermal charging rate limit; scarcity rent of the charging capacity of thermal storages and heat pumps	CHF/MW
storageTh_soc_dual(o)	Dual value of the thermal storage balance constraint; thermal storage value	CHF/MWh _{th}
storageTh_soc_limit _dual(o)	Dual value of the thermal storage energy capacity limit; scarcity rent of the thermal storage energy capacity	CHF/MWh _{th}
sum_generation_plant _energy_limited(i)	Annual upper generation limit of certain technologies	MWh
TES_decayrate(i)	Daily discharge rate of thermal storages	
th_sl(o)	Thermal storage level of households heating flexibly with heat pumps with 0 corresponding to the comfortable reference temperature (22°C)	MWh
trade_cost_dict (o)	Only used in a Switzerland-only configuration that belongs to a separate model variant; zero in standard runs and can be ignored	
V2G_charging_power_rate (i)	Max charging and discharging V2G power rate	MW
V2G_storage_capacity(i)	Max storage capacity of all V2G	MWh
weight_in_objective_fcn (i)	Weight of the individual subscenarios on the objective function	